

Department of Mathematics  
University of Arizona  
617 N Santa Rita Ave.  
Tucson, AZ 85721

williamchuang@arizona.edu

<https://williamchuang.github.io>

## EDUCATION

### University of Arizona

*M.S. in Mathematics (Expected)*

Spring 2025

(Graduate-level coursework completed under Ph.D. classification)

### San Francisco State University

*M.A. in Mathematics*

Spring 2022

**Thesis:** *The Hausdorff Dimension of Limit Sets of Well-Distributed Schottky Groups*

**Link:** <https://scholarworks.calstate.edu/downloads/xd07h079g>

**Advisor:** Dr. Chun-Kit Lai

### University of San Francisco

*B.S. in Mathematics, Minor in Computer Science*

Fall 2018

**Major GPA:** 3.88/4.00

Graduated with Honors

## RESEARCH INTERESTS

I am broadly interested in the interplay between mathematical physics, geometry, and the topology of data. My recent work focuses on:

- **Developing Quantum-Ready Transformer/Autoencoder Architectures from Geometric First Principles:** Constructed a novel framework wherein the dynamics of attention and encoding layers are governed by Möbius flows in  $\mathrm{PSL}(2, \mathbb{R})$ , derived directly from differential equations of the form  $\frac{dz}{dt} = \kappa(z - z_-)(z - z_+)$  associated with isometries. Each layer corresponds to the exponential of a Lie algebra element in  $\mathfrak{sl}(2, \mathbb{R})$ , naturally forming a one-parameter flow that preserves hyperbolic geometry. This system:
  - Enables exact quantization via Schrödinger-type evolution under unitary group representations;
  - Encodes model evolution as geodesic motion in curved latent spaces, improving interpretability and structure-preserving compression;
  - Outperforms existing quantum ML proposals by unifying classical geometry and quantum operator theory from first principles;
  - Provides a new foundation for quantum LLMs where activation and attention flows evolve by analytically integrable, symmetry-conserving dynamics.

- Analyzing the intrinsic factors and topological structures underlying self-attention architectures, aiming to understand how geometry and topology inform algorithmic design.
  - Developing theoretical frameworks in geometric data analysis, notably through an abstract definition of curvature for novel data representations.
  - Applying hyperbolic geometry and Kleinian groups to enhance neural network performance and security—specifically:
    - (1) Introducing global symmetries—recognized as a parallel to Poincaré’s monodromy—into neural networks, particularly in autoencoders and transformers, by leveraging group actions on these networks. This extends monodromy beyond its classical role in Fuchsian differential equations to all intermediate solution paths in a neural network, structuring the entire optimization landscape through group-theoretic mappings.
    - (2) Strengthening data encryption and decryption,
    - (3) Devising new approaches for deriving neural networks that can be composed or edited mathematically based on previously trained models,
    - (4) Dynamically modifying energy landscapes on a global scale to reduce computational costs during training by sequentially mapping model parameters to configurations that systematically decrease the loss function’s output.
  - Investigating linear combinations, cyclic groups, and orbifolds of feedforward neural networks with restricted codomains, leveraging autoencoders composed of feedforward networks and their inverses to achieve deeper model interpretability.
  - Constructing a rigorous interaction theory by integrating:
    - (1) The Callan–Symanzik equation and stochastically motivated measures,
    - (2) Motives and Hopf algebras within a renormalization group (RG) framework,
 to advance the mathematical understanding of deep neural networks.
- 

## ORIGINAL CONTRIBUTIONS AND HISTORICAL FIRSTS

The following research contributions represent original discoveries, each constituting a historical first in its respective domain, based on available literature and records:

- (1) **Redefining Core Algebra in Neural Networks Using Hyperbolic Geometry:**  
 This directly unlocks model orbit mobility under group actions—achieving a level of control and transformation that Marvin Minsky famously proved to be computationally intractable, even NP-hard, in the classical perceptron model (cf. *Perceptrons*, 1988, with Papert).
  - *Historical First:* No one has previously published a fully geometrically consistent, hyperbolically grounded framework in deep learning that enables this level of group-based model orbit mobility.
  - *Recent Extension:* This foundational structure has now been elevated to support an infinite family of mathematically consistent geometric methods—each preserving

the expressive power of conventional Euclidean architectures, including the ability to represent negative-valued parameters and outputs. These constructions provide a unified, invariant framework for building deep neural networks, transformers, autoencoders, and any future architectures constructed from conventional deep learning primitives. Each variant within this framework may be optimally suited to different use cases depending on geometric, topological, analytical, algebraic, or statistical/probabilistic considerations—opening a new paradigm for principled architecture design in intelligent systems, first-principles quantum neural networks, and post-machine-learning and post-quantum cryptographic applications.

- (2) **First Quantum-Ready Transformer Architecture Built from Geometric Principles:** Developed a quantum transformer and autoencoder design where each layer corresponds to Möbius transformations generated by Lie algebra symmetries. This structure naturally evolves through Schrödinger-type dynamics and is built from first principles rather than heuristic circuit designs.
- (3) **Monodromy of Neural Networks:** Introduced the idea of tracking how a deep learning model evolves globally as its parameters move in loops—extending the classical concept of monodromy to the training dynamics of modern AI models.
- (4) **Topological Entropy as a Cryptographic Hardness Parameter:** Embedded the Hausdorff dimension  $\delta$  into cryptographic design as a tunable constant that governs complexity, obfuscation, and resistance against both classical and quantum attacks.
- (5) **Neural Encryption via the Holographic Principle:** Designed a cryptographic neural architecture grounded in the holographic principle, where data and parameters are embedded in curved latent spaces. Established a formal bulk–boundary correspondence:
  - Input tokens  $\leftrightarrow$  boundary observables,
  - Transformer layers  $\leftrightarrow$  bulk fields in latent space,
  - Weight dynamics  $\leftrightarrow$  renormalization flow,
  - Cipher inversion  $\leftrightarrow$  boundary reconstruction from bulk geometry,
  - Semantic representation  $\leftrightarrow$  automorphic kernel functions.

The encryption mechanism requires full latent structure for decryption, making partial inversion equivalent to reconstructing bulk geometry from boundary data—thus encoding strong cryptographic hardness rooted in holographic duality.

- (6) **AdS/CFT-Inspired Neural Cryptosystem:** Explicitly modeled a neural architecture on the AdS/CFT correspondence. The bulk geometry was implemented using hyperbolic latent space ( $\mathbb{H}^{n+1}$ ), with transformer layers representing bulk fields and outputs interpreted as boundary CFT observables. The system reaches a theoretical security tier informally comparable to an “S++ class” in analogy with NSA-style classification schemes, due to its reliance on unsolved quantum gravity dualities and automorphic obfuscation. Constructed a structural map:
  - Tokens  $\leftrightarrow$  CFT operators on the boundary,
  - Transformer depth  $\leftrightarrow$  radial AdS coordinate,
  - Parameter flow  $\leftrightarrow$  RG flow in bulk field theory,
  - Model inversion  $\leftrightarrow$  AdS/CFT duality inversion,

- Compression and encoding  $\leftrightarrow$  minimal surfaces (Ryu–Takayanagi formula).
- *Implementation Clarification:* Importantly, this approach does not assume our physical universe is Anti-de Sitter. Instead, the AdS “bulk” is rendered as a computational geometry within the model—analogous to how physics engines simulate curved spaces in virtual environments. The duality is symbolically encoded in the model’s tensor structure, with geodesic kernels and partition-function-based attention. Thus, the AdS/CFT framework is not a metaphysical claim but an engineered duality, implemented purely through transformer design and hyperbolic geometry, and fully realizable on conventional hardware.

This cryptosystem achieves maximal obfuscation by leveraging the intractability of reconstructing bulk geometry from boundary data—an encryption-hardness analog of AdS/CFT inversion.

- (7) **Embedding Langlands Duality in Neural Cryptographic Design:** Proposed a GPT-style model architecture that encodes Langlands duality principles at the structural level. No prior work in AI or cryptography formalized this correspondence.
- (8) **First Use of Automorphic Forms in Attention Mechanisms:** Augmented the attention mechanism in transformers using automorphic functions, linking neural language representations to deep structures from harmonic analysis.
- (9) **Loss Landscape Control via Lorentz Transformations:** Discovered that Lorentz transformations can systematically reshape the energy landscape of a model, offering a new, geometry-based alternative to conventional gradient descent for training.
- (10) **Modular Spectral Indistinguishability Assumption (MSIA) – Post-Quantum Cryptographic Primitive:** Introduced the MSIA framework, a novel cryptographic assumption built on modular spectral factorization, symbolic dynamics, and modular representation theory. The core challenge is the inversion or factorization of structured zeta functions over finite fields:

$$\zeta_p(z) = \frac{1}{\det(I - zA^{(p)})}, \quad A^{(p)} \in \mathbb{F}_p^{n \times n}.$$

The structure embeds rich combinatorial and algebraic constraints, including projective indecomposable modules and symbolic Brauer classes.

- *Security Model:* Achieves theoretical hardness analogous to NP-hard and #P-hard problems under both classical and quantum adversarial models. The framework supports trapdoor-based encryption with up to  $100\times$  parameter amplification, enabling layered obfuscation mechanisms.
- *Classification:* Under informal cryptographic hardness analogies (e.g., C, B, A, S, S+), MSIA with appropriate scaling can reach the S+ level, and with additional symbolic or hybrid obfuscation layers, may approach an S++-class cryptosystem in terms of theoretical resistance to inversion. These terms are used metaphorically to denote exceptional post-quantum intractability and are not tied to formal government classification levels.

- *Historical First:* First known use of modular trace zeta functions and symbolic module structures to construct post-quantum cryptographic hardness assumptions within a structured, spectral-theoretic framework.
- 

## OTHER INDEPENDENT PROJECTS

### University of Arizona

- **Independent Study: Real and Complex Analysis, and Applications of Hyperbolic Geometry**

With Prof. David Glickenstein, Fall 2023

### University of Arizona

- **RTG Project: Scaling Factors of Self-Attention Weights in Transformers**

With Prof. Ning Hao, Fall 2023

### San Francisco State University

- **Computation of the Hausdorff Dimension of Limit Sets of Schottky Groups**

With Dr. Chun-Kit Lai, June 2021 – May 2022

### San Francisco State University

- **Independent Study: Prime Geodesic Theorem and Limit Sets of Schottky Groups**

January 2021 – May 2021

Wrote a summary of the modern proof with an emphasis on growth rates based on the Hausdorff dimension of the associated limit set.

*Advisor:* Dr. Chun-Kit Lai

### San Francisco State University

- **Topology Project: A Study on Fundamental Groups**

September 2020 – December 2020

*Advisor:* Dr. Emily Clader

### San Francisco State University

- **Independent Study: Hom-Polytopes**

September 2019 – December 2019

- **Combinatorics Project: Simplicial Complexes**

January 2019 – May 2019

*Advisor:* Dr. Joseph Gubeladze

### University of San Francisco

- **Independent Study: Prime Number Theorem**

January 2018 – May 2018

*Advisor:* Dr. Paul Zeitz

### Pennsylvania State University–University Park

- **Functional Analysis Project: Hardy's Proof of Uniform Distribution**

January 2018 – May 2018

- **Independent Study: Reading “Lecture Notes on Functional Analysis: With Applications to Linear Partial Differential Equations”**

January 2018 – May 2018

*Advisors:* Dr. Sergei Tabachnikov and Dr. Moisey Guysinsky

#### **Pennsylvania State University–University Park**

- **Topology Project: Solving the (9, 8, 4, 3, 7)-Linkage Problem**

January 2018 – May 2018

- **Topology Final Project: Conway’s Basic Theorem**

September 2017 – December 2017

*Advisor:* Dr. Sergei Tabachnikov

#### **University of San Francisco**

- **Capstone Project: Graph Theory for an Inverted-Index-Based Search Engine**

January 2018 – May 2018

*Advisor:* Dr. Chris Bryan

#### **University of San Francisco**

- **Capstone Project: Applying the Method of Steepest Descent and Cauchy Contour Integrals to the Fisher Exact Test**

January 2018 – May 2018

*Advisor:* Dr. Xuemei Chen

#### **University of San Francisco**

- **Research Assistant**

August 2016 – May 2017

Assisted with lecture notes for MSAN 504 (Review of Probability and Statistics).

*Advisor:* Dr. Jeff Hamrick

#### **University of San Francisco**

- **Capstone Project: Implementing Dijkstra’s Algorithm Applications**

Spring 2016

- **Summer Research Project: Therapeutic Video Games for Patients with Disabilities**

June 2016 – September 2016

- **Interpreting Deep Neural Networks**

Fall 2016

Explored causal structures within deep networks to map them onto symbolic graphs, and investigated methods to initialize models from human-written code.

Read causal inference research by Prof. David Galles and Judea Pearl.

*Advisor:* Dr. David Galles

#### **PRE-BACCALAUREATE INDEPENDENT PROJECTS**

#### **National Taiwan University**

- **Research Student at LeCosPA**

September 2011 – May 2013

Presented various topics in weekly meetings and seminars, including:

- Bremsstrahlung and Cherenkov radiation
- Topological quantum field theory and 2+1D quantum gravity via Chern-Simons terms
- Cosmological constant, vacuum structure, and vacuum energy
- Radiation from moving mirrors and black holes (Schwinger mechanism, Casimir effect, Hawking/Unruh effects)
- Potential carbon-free energy sources via low-energy nuclear reactions
- Metamaterials and analog models of gravity
- Instability of Anti-de Sitter space
- Induced gravity, Coleman-Weinberg–Witten theorem on Lorentz violation, AdS/CFT correspondence
- Quantum information, holographic turbulence, AdS/CMT, sonoluminescence
- Holographic renormalization group flow and Ricci flow
- Background-independent spin foam models and Regge calculus

*Advisor:* Dr. Pisin Chen

**National Taiwan University**

- **Kontsevich–Soibelman Wall-Crossing Formula for Mathematical QFT**

January 2012 – May 2012

- **Conformal Bootstrap Methods for the 3D Ising Model**

2011

*Advisor:* Dr. Heng-Yu Chen

**National Taiwan University**

- **A Study on the Lee–Yang Theorem and Riemann Zeta Function in Statistical Mechanics**

January 2012 – May 2012

*Advisor:* Dr. Ning-Ning Pang

**National Taiwan University**

- **Dark Energy Problem via Modified Gravity**

September 2010 – May 2011

Focused on the equivalence of Einstein frames and conformal mappings in scalar-tensor theory.

*Advisor:* Dr. Je-An Gu

---

WORK EXPERIENCE (TEACHING & RESEARCH)

**University of Arizona**

- Graduate Teaching Assistant, MATH 112 (College Algebra), Section 33 – Fall 2022
- Graduate Teaching Assistant, MATH 112 (College Algebra), Sections 12 & 18 – Spring 2023

- Graduate Teaching Assistant, MATH 112 (College Algebra), Section 13 – Fall 2023
- Grader, MATH 112 (College Algebra), Sections 9, 13 & 20 – Fall 2023
- Tutor, MATH 129 (Calculus II) – Fall 2023
- Grader, MATH 129 (Calculus II) Final Exam – Fall 2023
- Grader, MATH 122B/125 (Calculus I) Common Final Exam – Fall 2023
- Graduate Teaching Assistant, MATH 112 (College Algebra), Sections 101, 102, 201, 202, 401 & 402 – Spring 2024
- Graduate Teaching Assistant, MATH 125 (Calculus I), Section 001 – Fall 2024
- Graduate Teaching Assistant, MATH 112 (College Algebra), Sections 103 & 203 – Spring 2025
- **Teaching Mentors & Advisors:** Mitchell Wilson, Tina Deemer, Catherine Yslas, Oussama Ben Said, Tynan Lazarus, and Prof. David Glickenstein

### **San Francisco State University**

- Graduate Teaching Assistant, Calculus – Spring 2022
- Grader, MATH 227 [05] (Calculus II)
- Instructor, MATH 226 [38] (Calculus I) – Fourth-hour component of MATH 226 [37]
- Instructor, MATH 227 [06] (Calculus II) – Fourth-hour component of MATH 227 [05]
- Instructor, MATH 227 [36] (Calculus II) – Fourth-hour component of MATH 227 [35]
- **Advisors:** Prof. Kim Seashore, Prof. Shandy Hauk, and Prof. Eric Hsu

## OTHER ACADEMIC EXPERIENCE

### **San Francisco State University**

- Graduate Teaching Assistant, Pre-Calculus – Fall 2019
- Advisor: Prof. Kim Seashore

### **University of San Francisco**

- San Francisco Math Circle – Fall 2016
- Advisor: Prof. Paul Zeitz

### **National Dong Hwa University**

- Undergraduate Research Assistant – Spring 2010 Hired and advised by Prof. Cheng-Pang Liu
- Tutor of Calculus and General Physics – August 2008 to December 2009 Hired by the NDHU Department of Physics

---

## AWARDS AND HONORS

- **Nominated for MSRI Summer Graduate School on Metric Geometry and Geometric Analysis**  
University of Oxford (UK), Fall 2021
- **Dean's Honor Roll**  
University of San Francisco, Spring 2018



- **Mathematics Advanced Study Scholarship and Internal Scholarship (MASS Program)**  
The Pennsylvania State University–University Park, Fall 2017  
(Covered tuition and fees)
  - **Dean’s Honor Roll**  
University of San Francisco, Spring 2015, Fall 2016, and Spring 2017
  - **Pi Mu Epsilon Honor Society**  
University of San Francisco
  - **Admitted to the Summer School on Symmetry in Mathematics and Physics**  
National Taiwan University, Summer 2012
  - **Admitted to Prof. Anthony Zee’s Quantum Field Theory Course**  
Institute of Physics, Academia Sinica, February 2012
  - **Admitted to the 1st LeCosPA Symposium: Towards Ultimate Understanding of the Universe**  
National Taiwan University, February 2012  
[\[Link\]](#)
  - **Admitted to the 2nd International Workshop on Dark Matter, Dark Energy, and Matter-Antimatter Asymmetry**  
National Tsing Hua University, Winter 2010  
[\[Link\]](#)
  - **Admitted to the Summer School for Theoretical Physics**  
National Tsing Hua University, Summer 2009
  - **President’s List**  
National Dong Hwa University, March 2008, November 2008, March 2009, March 2010
- 

#### CERTIFICATES

- **Protecting God’s Children Online Awareness 4.0**  
Order of Malta, Western Association – March 7, 2025
- **Safety Preparedness Training**  
The University of Arizona, Employee Development, Growth, and Engagement  
December 8, 2023
- **Information Security Awareness Certification**  
The University of Arizona, Employee Development, Growth, and Engagement  
August 27, 2023
- **MASS Program Completion**  
Completed all requirements for the 2017 Mathematics Advanced Study Semesters program at The Pennsylvania State University
- **Recognition of Service Award**  
ACM Special Interest Group on Management of Data (SIGMOD) – 2016
- **Tackling the Challenges of Big Data**  
Online program developed by the faculty of the MIT Computer Science and Artificial

## SERVICE TO THE COMMUNITY

### **Early Service Initiatives (1998 – 2003).**

- 1998 – 2003 – Altar Server (Mass Assistant) at Fu Jen Catholic University, assisting at liturgical celebrations attended by university professors and students.
- 2000 – First engagement with service learning at Boston College, accompanying my father, a seed professor for Fu Jen Catholic University's Service Learning Program. Taught elementary arithmetic to children from minority communities in Boston.

### **High School Service (2000 – 2006).**

- 2000 – 2006 – Served a minimum of 15 hours per year at local hospitals in Taipei through St. Ignatius High School, assisting:
  - Elderly patients, feeding them spoon by spoon.
  - Children with cerebral palsy and other disabilities, ensuring they received meals with care and dignity.

### **University and Adult Service (2017 – Present).**

- 2017 (Spring) – Teaching Assistant for Mathematics Enrichment Program at the University of San Francisco (USF), providing academic support and instructional assistance to students.
- 2023 (October 16 – Present) – Volunteer at the University of Arizona Tucson Math Circle, primarily maintaining the website to support outreach and mathematical education.
- 2025 (February 21) – Began volunteering at Poverello House (Tucson, AZ), assisting the homeless community (4-hour shift).
- 2025 (March 29, scheduled) – Continuing volunteer service at Poverello House (Tucson).
- 2025 (May 18, upcoming) – Volunteering at a Blood Drive (9 AM – 12 PM) organized by the Knights of Columbus at St. Thomas the Apostle Parish (Tucson, AZ).

### **Future Service Commitment.**

- Following my graduation from the University of Arizona (Spring 2025), I plan to increase my volunteer engagement with the Order of Malta, Poverello House, the Knights of Columbus, Catholic Action in San Diego, and other service initiatives.

### **Faith Formation and Study.**

- Regular recipient and self-study of Opus Dei's publications, integrating its spiritual insights into daily life, professional service, and ethical decision-making.
- Active engagement with Catholic Action in San Diego, remotely supporting its evangelization efforts and operations through prayer and spiritual solidarity.

- Ongoing study of books published by the Order of Malta and its members, including the Catholic Bible, prayer books, and *Minutes with the Catechism*, deepening my understanding of faith and service.
- 

## SKILLS

- **Problem Solving and Adaptability:** Demonstrated ability to learn new skills quickly.
- **Programming Languages:** C/C++, Python, R, Java, Lisp, Shell Script, Sed, Awk, L<sup>A</sup>T<sub>E</sub>X, Mathematica
- **Libraries and Packages:** PyTorch, Lightning, NumPy, Pandas, scikit-learn, Matplotlib, Ogre3D
- Designing algorithms to generate examples for theoretical research in mathematics, physics, statistics, and computer science